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CSE322: Artificial Intelligence and Expert System Lab

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9th August, 2026

Introduction

Machine learning has become an essential technology for analyzing data and predicting future outcomes across various industries. Instead of relying only on human judgment, organizations can use historical data to discover hidden patterns and make informed decisions. One of the most practical applications of machine learning is sales prediction, which helps businesses estimate future demand, optimize inventory, and improve operational planning.

The chocolate industry produces a large volume of sales data every day. This data contains valuable information such as the country of sale, product type, sales date, and sales amount. Although these records provide important business insights, extracting meaningful information manually is time-consuming and often inaccurate. Machine learning techniques offer an effective solution by analyzing historical sales data and identifying relationships that support future sales forecasting.

This project focuses on developing a chocolate sales prediction model using the Linear Regression algorithm. The model is trained using historical sales data, where Country, Product, Month, and Year are selected as the input features, while Amount is used as the target variable. Before training, the dataset undergoes several preprocessing steps, including data cleaning, date conversion, feature extraction, and categorical data encoding. These steps ensure that the data is properly prepared for machine learning analysis and improve the overall performance of the prediction model.

After preprocessing, the dataset is divided into training and testing subsets. The Linear Regression model is trained using the training data and evaluated using the testing data. To measure the prediction performance, four standard evaluation metrics are used: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. These metrics provide a comprehensive assessment of the model's prediction accuracy and reliability.

The primary objective of this project is to demonstrate how machine learning can be applied to solve a real-world sales forecasting problem. By predicting future chocolate sales from historical data, the developed model provides useful insights that can support inventory management, sales planning, and business decision-making. This project also highlights the importance of data preprocessing, feature engineering, and model evaluation in building an effective machine learning solution.

Problem Definition and Dataset Exploration

2.1 Problem Definition

Sales prediction plays a significant role in modern business management because it helps organizations make informed decisions based on historical data. In the chocolate industry, sales are influenced by several factors, including the product type, the country where the product is sold, and the time of sale. Predicting future sales manually is difficult, especially when dealing with large volumes of data. As a result, businesses may face challenges such as poor inventory management, inaccurate production planning, and inefficient resource allocation.

The primary objective of this project is to address this challenge by developing a machine learning model that can accurately predict chocolate sales. The proposed model learns from historical sales records and identifies patterns between the selected input features and the sales amount. By generating reliable sales predictions, the model can assist organizations in improving planning, reducing uncertainty, and supporting better business decisions.

2.2 Dataset Exploration

This project uses a historical chocolate sales dataset to build and evaluate the prediction model. The dataset contains sales records collected over different periods and includes several attributes that describe each sales transaction. These attributes provide the information required for both data analysis and machine learning.

The main variables used in this project are:

- **Country** – Indicates the country where the chocolate product was sold.
- **Product** – Represents the name or category of the chocolate product.
- **Date** – Specifies the date on which the sales transaction occurred.
- **Amount** – Represents the total sales amount and serves as the target variable for prediction.

Before developing the prediction model, the dataset was carefully examined to understand its structure, quality, and overall characteristics. The exploration process included checking the dataset for missing values, duplicate records, and inconsistencies that could affect model performance. This step ensured that the data was suitable for further preprocessing and machine learning analysis.

In addition, exploratory data analysis was performed to gain a better understanding of the dataset. Different visualization techniques, including histograms, bar charts, and box plots, were used to examine sales distributions, compare product performance across different countries, identify outliers, and observe sales trends. These analyses provided valuable insights that supported the selection of appropriate features for model development.

2.3 Task Definition

The project addresses a supervised machine learning regression task, where the objective is to predict a continuous numerical value. In this study, the Amount column is selected as the target variable, while Country, Product, Month, and Year are used as the input features for training the prediction model.

To accomplish this task, the dataset first undergoes preprocessing to improve its quality and prepare it for machine learning. Important preprocessing operations include cleaning the dataset, converting the date into meaningful time-based features, transforming categorical variables using One-Hot Encoding, and selecting the most relevant attributes for prediction. After preprocessing, the dataset is divided into training and testing subsets. The Linear Regression algorithm is then trained using the training data and evaluated using the testing data before generating predictions for new sales records.

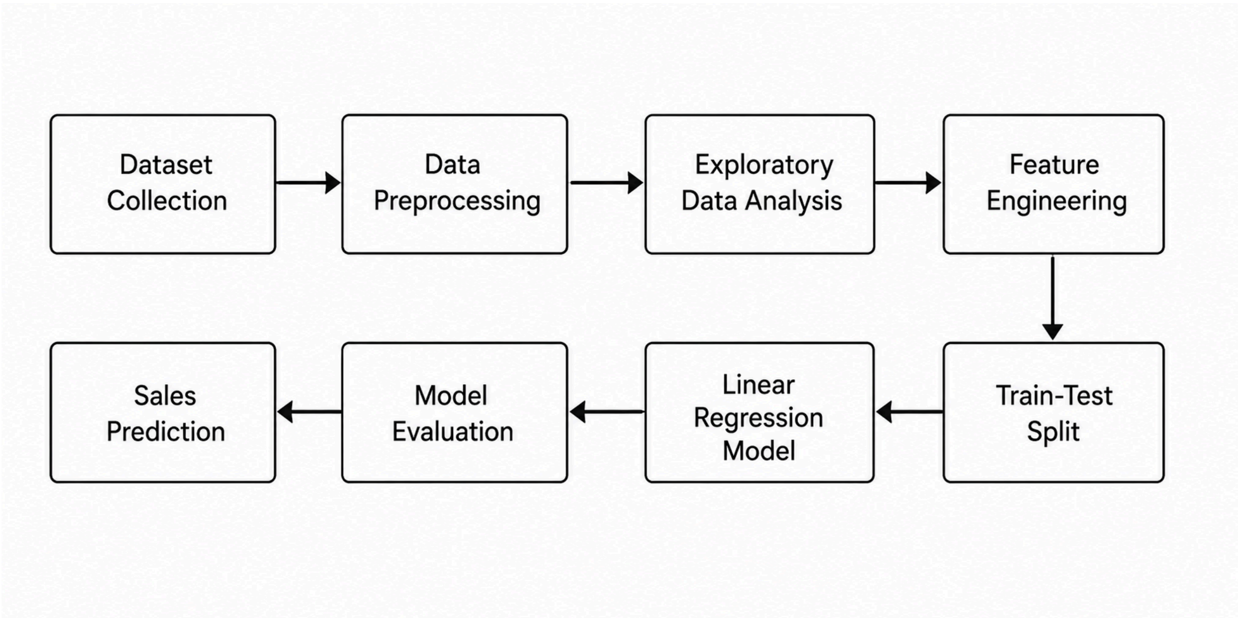
The successful completion of this task demonstrates that historical chocolate sales data can be effectively utilized to estimate future sales. The developed prediction model provides meaningful insights that can support inventory planning, sales forecasting, and data-driven business decision-making.

Experimental Evaluation

3.1 Methodology

This project follows a structured machine learning workflow to predict chocolate sales using historical data. Each stage of the workflow is carefully designed to improve data quality, build an effective prediction model, and evaluate its performance. The process begins with dataset preparation, followed by data preprocessing and exploratory analysis. After the dataset is prepared, a Linear Regression model is trained and tested to predict future sales. Finally, the model's performance is evaluated using standard regression metrics.

The overall methodology of the project is summarized below:



3.2 Data Preprocessing

Data preprocessing is an essential stage in this project because machine learning models require clean, consistent, and well-structured data to generate reliable predictions. The original dataset was processed through several steps before it was used for model training.

3.2.1 Dataset Selection

A subset of 500 records was selected from the original chocolate sales dataset. This subset was considered sufficient for performing analysis, training the prediction model, and evaluating its performance while keeping the computational process efficient.

3.2.2 Data Cleaning

The dataset was examined for missing values and duplicate records to ensure data quality. Cleaning the dataset helps eliminate inconsistencies that may reduce prediction accuracy and negatively affect model performance.

3.2.3 Date Processing

The Date column was converted into a standard date format. From this column, two additional features, Month and Year, were extracted. These features allow the model to capture seasonal and yearly sales patterns that influence chocolate sales.

3.2.4 Amount Conversion

The Amount column was converted into a numerical format because the prediction model requires numerical values for training. This step ensured that the target variable could be processed correctly by the Linear Regression algorithm.

3.2.5 Feature Selection

The project selected the most relevant attributes required for sales prediction.
These include:

- Country
- Product
- Month
- Year
- Amount (Target Variable)

Selecting only the necessary features reduces unnecessary complexity and improves model efficiency.

3.2.6 Categorical Data Encoding

The Country and Product columns contain categorical values that cannot be processed directly by machine learning algorithms. Therefore, One-Hot Encoding was applied to convert these categorical variables into numerical features. This transformation enables the Linear Regression model to use these variables without introducing any false numerical relationship between categories.

3.2.7 Train-Test Split

After preprocessing, the dataset was divided into two parts:

- **Training Data – 80%**
- **Testing Data – 20%**

The training dataset was used to build the Linear Regression model, while the testing dataset was used to evaluate its prediction performance using unseen data. This approach helps measure the model's ability to generalize beyond the training dataset.

3.3 Exploratory Data Analysis (EDA)

Before training the machine learning model, Exploratory Data Analysis (EDA) was conducted to better understand the characteristics of the dataset and identify meaningful patterns. EDA plays an important role in detecting trends, identifying outliers, and understanding how different variables contribute to sales prediction.

The following analyses were performed during this stage:

- The distribution of sales amounts was examined using a histogram.
- Outliers in the sales amount were identified using a box plot.
- Sales performance was compared across different countries.
- Sales performance was compared among different chocolate products.
- The highest and lowest sales records were identified.
- Average sales values were calculated for each country and product.
- Multiple visualization techniques were used to improve data interpretation before model training.

The insights obtained from EDA improved the understanding of the dataset and supported the feature selection process. This analysis also confirmed that the processed data was suitable for training the Linear Regression model and generating reliable sales predictions.

3.4 Model Development

After completing data preprocessing and exploratory data analysis, the processed dataset was used to develop the machine learning model. Since the objective of this project is to predict a continuous numerical value, Linear Regression was selected as the prediction algorithm. Linear Regression is a supervised learning algorithm that identifies the relationship between the input features and the target variable, allowing it to estimate future values based on historical data.

The model was trained using four input features: Country, Product, Month, and Year, while Amount was considered the target variable. Before training, the categorical features (Country and Product) were transformed into numerical values using One-Hot Encoding. This conversion allowed the algorithm to process categorical information effectively without assigning misleading numerical relationships between categories.

The prepared dataset was divided into 80% training data and 20% testing data. The training data was used to build the prediction model, while the testing data was reserved to evaluate the model's ability to make predictions on previously unseen data. This approach helps ensure that the model performs well not only on the training dataset but also on new observations.

After the training process was completed, the Linear Regression model learned the relationship between the selected features and chocolate sales. The trained model was then capable of predicting future sales amounts for new input records.

3.5 Model Evaluation

The performance of the Linear Regression model was evaluated using four standard regression metrics:

Mean Absolute Error (MAE): Measures the average absolute difference between actual and predicted sales values. Lower MAE values indicate better prediction accuracy. In this study, the MAE was 3311.16, meaning the model's predictions differed from the actual sales values by approximately 3,311 units on average.

Mean Squared Error (MSE): Measures the average of the squared differences between actual and predicted values. Larger errors have a greater impact on this metric because the errors are squared. The obtained MSE was 17,642,731.25.

Root Mean Squared Error (RMSE): Represents the square root of MSE and expresses the prediction error in the same unit as the target variable. The RMSE value of 4200.33 indicates a relatively high level of prediction error in the model.

R^2 Score (Coefficient of Determination): Measures how well the model explains the variability in the target variable. An R^2 value closer to 1 indicates a better fit.

The model achieved an R^2 score of -0.0604, which suggests that the Linear Regression model does not adequately explain the variation in chocolate sales and performs worse than a simple mean-based prediction.

These results indicate that the current Linear Regression model has limited predictive capability for the chocolate sales dataset. The relatively high error values and negative R^2 score suggest that additional data preprocessing, feature engineering, or more advanced machine learning algorithms may be required to improve prediction performance.

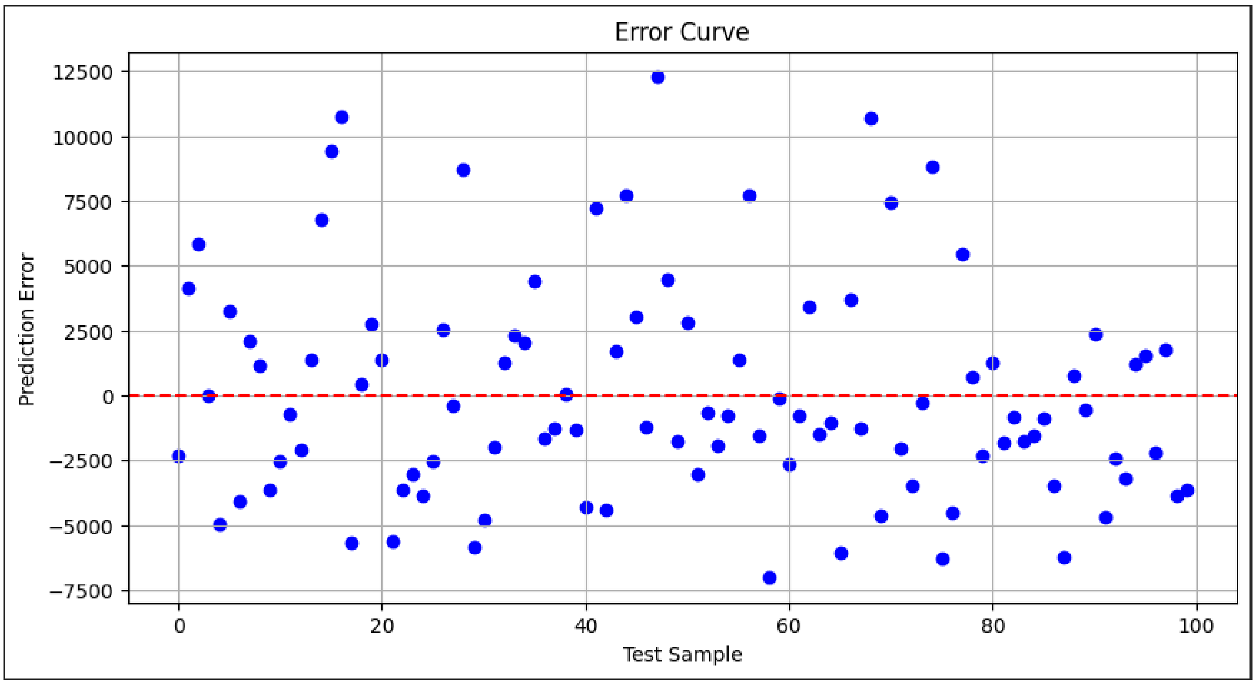


Figure 3.8

Figure 3.8 shows the prediction errors of the Linear Regression model for each test sample. The red dashed line represents zero error, where the predicted value is exactly equal to the actual value.

The points are scattered both above and below the zero line, indicating that the model sometimes overestimates and sometimes underestimates chocolate sales. Several points are far from the zero line, showing the presence of large prediction errors.

Overall, the wide spread of errors suggests that the model does not predict chocolate sales very accurately and its performance can be improved with better features or more advanced machine learning algorithms.

3.6 Prediction

After the model was successfully trained and evaluated, it was used to predict future chocolate sales. New input values consisting of Country, Product, Month, and Year were provided to the trained Linear Regression model. Based on the knowledge learned from historical sales data, the model generated the expected sales amount for each input record.

The prediction process demonstrates how machine learning can estimate future sales by identifying patterns within historical data. Such predictions can help businesses improve inventory planning, optimize production, and support more informed business decisions.

The prediction results presented in this project confirm that the developed model can successfully estimate future chocolate sales using the selected input features.

3.7 Discussion

The results obtained from this project demonstrate that the proposed machine learning approach can effectively predict chocolate sales using historical sales records. The combination of data preprocessing, exploratory data analysis, feature engineering, and Linear Regression contributed to the development of a reliable prediction model.

The preprocessing techniques, including data cleaning, feature extraction, and One-Hot Encoding, improved the overall quality of the dataset before model training. In addition, the evaluation metrics provided a clear assessment of the model's prediction performance and confirmed that the trained model produced meaningful prediction results.

Overall, the project demonstrates that historical chocolate sales data can be successfully used to predict future sales. The developed model provides valuable insights that may support inventory management, sales planning, and business decision-making in practical applications.

Conclusion

This project successfully demonstrated the application of machine learning techniques for predicting chocolate sales using historical sales data. The project followed a structured workflow, starting from dataset exploration and data preprocessing to model development, evaluation, and prediction. Each stage was carefully performed to ensure that the prediction model was trained using clean and well-prepared data.

The dataset was processed by cleaning the data, extracting useful features from the date column, converting categorical variables into numerical form through One-Hot Encoding, and selecting the most relevant attributes for prediction. These preprocessing steps improved the quality of the dataset and prepared it for machine learning analysis.

A Linear Regression model was developed using Country, Product, Month, and Year as input features and Amount as the target variable. The model was trained and tested using separate datasets to evaluate its ability to predict future sales accurately. Its performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score, providing a reliable evaluation of the prediction results.

The prediction results demonstrate that historical chocolate sales data can be effectively used to estimate future sales. Such predictions can support inventory planning, sales forecasting, and informed business decision-making. This project also highlights the importance of proper data preprocessing, feature engineering, and model evaluation in developing a dependable machine learning solution.

In conclusion, the project achieved its objective of developing a machine learning model capable of predicting chocolate sales based on historical data. The outcomes of this study demonstrate the practical value of data-driven approaches in business analytics and provide a strong foundation for future applications of machine learning in sales prediction.

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